Student Enrollment & Course Management System

Project Documentation Report

Submitted by: _____

Department: _____

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1. Introduction

Educational institutions handle thousands of student records, course offerings and enrollment requests every semester. When these tasks are done manually or through scattered spreadsheets, the result is duplicate entries, seat-allocation conflicts, delayed approvals and poor visibility for administrators.

The **Student Enrollment & Course Management System (SECMS)** is a web-based application that centralises student registration, course creation, seat management, enrollment approval and reporting in one secure platform. Students can browse courses and enroll online, faculty can view class lists, and administrators can control the complete academic workflow.

Abstract

This report describes the analysis, design and testing of SECMS. The system follows a three-tier architecture, uses a relational database with role-based access control, and automatically validates prerequisites, seat availability and timetable clashes before confirming an enrollment.

2. Problem Statement & Objectives

2.1 Problem Statement

Existing manual enrollment processes suffer from long queues, paper-based forms, human errors in data entry, no real-time seat tracking and difficulty in generating reports. Students often do not know which courses are open, and administrators cannot quickly detect over-enrollment or timetable conflicts.

2.2 Objectives

- Provide online student registration and secure login for every user role.
- Allow administrators to create, update and retire courses with capacity limits.
- Automate enrollment with prerequisite, seat and timetable-clash validation.
- Give faculty instant access to class lists and attendance-ready rosters.
- Generate enrollment, seat-utilisation and student-history reports.
- Send automatic email/SMS notifications for approvals, waitlists and drops.

2.3 Scope

The system covers the complete enrollment lifecycle: registration, course search, enrollment, approval, waitlisting, dropping and reporting. Fee payment, examination management and hostel allocation are outside the current scope but the design allows integration later.

2.4 Stakeholders

Stakeholder	Interest
Students	Easy course search and quick enrollment
Faculty	Accurate class lists and course details
Administrators	Control, approvals and reporting
Management	Data-driven decisions on capacity and demand

3. System Requirements

3.1 Functional Requirements

ID	Requirement	Priority
FR1	User registration, login, logout and password reset	High
FR2	Admin can add, edit, deactivate courses and set seat capacity	High
FR3	Student can search courses by code, name, department or semester	High
FR4	Student can enroll in and drop courses within the enrollment window	High
FR5	System checks prerequisites, seats and timetable clashes	High
FR6	Automatic waitlist and promotion when a seat becomes free	Medium
FR7	Faculty can view enrolled students for assigned courses	Medium
FR8	Reports export to PDF and Excel	Medium

3.2 Non-Functional Requirements

- **Security:** password hashing, role-based access control, HTTPS.
- **Performance:** pages load within 3 seconds; supports 1,000 concurrent users.
- **Availability:** 99.5% uptime, especially during enrollment weeks.
- **Usability:** responsive interface that works on mobile and desktop.
- **Scalability:** database and server can scale as student count grows.

3.3 Hardware & Software

Category	Specification
Server	4-core CPU, 8 GB RAM, 100 GB storage
Frontend	HTML5, CSS3, JavaScript / React
Backend	Node.js / Java Spring Boot / Python Django
Database	MySQL 8 or PostgreSQL 15
Client	Any modern browser

4. System Architecture

SECMS uses a **three-tier architecture**, separating the user interface, business logic and data storage. This makes the system easier to maintain, secure and scale.

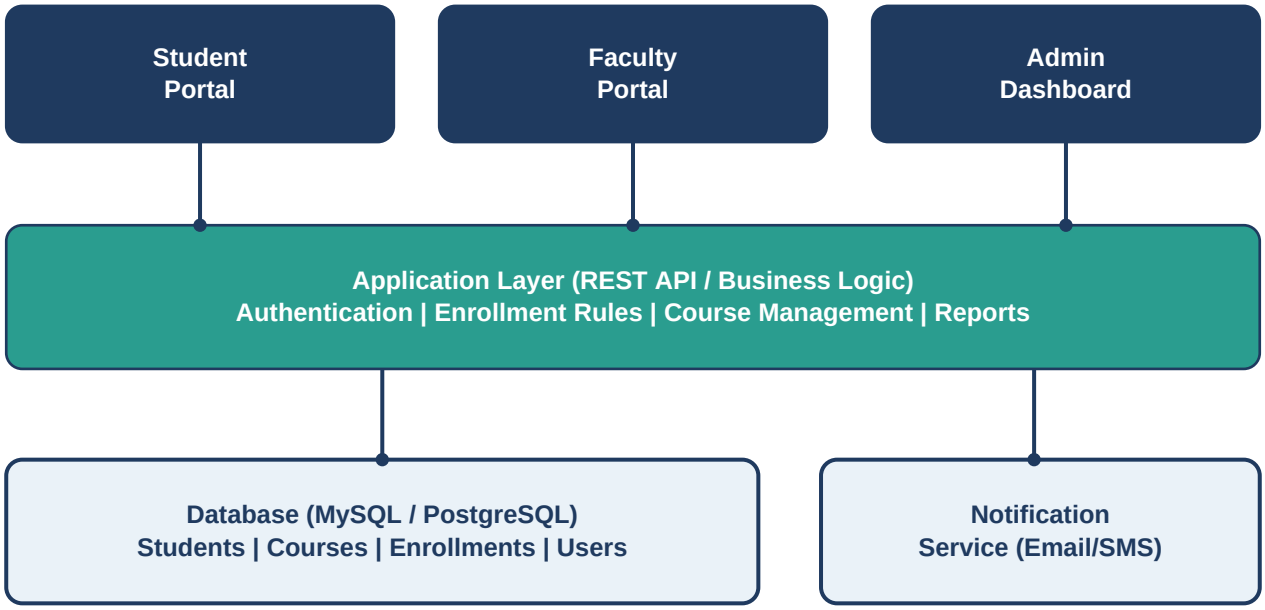

Figure 1: Three-tier architecture of the Student Enrollment & Course Management System

Layer Responsibilities

Layer	Responsibility
Presentation	Role-specific portals for students, faculty and admins; form validation and responsive display.
Application	REST APIs, authentication, enrollment rules engine, course logic and report generation.
Data	Persistent storage of users, courses, enrollments and audit logs with backups.
Notification	Sends confirmation, waitlist and reminder messages through email or SMS.

Communication between the client and server uses JSON over HTTPS. Every request carries a token that identifies the user and role, so unauthorised actions are rejected at the API level.

5. Modules Description

Module	Description	Key Features
User Management	Handles registration, authentication and roles.	Login, roles, password reset, profile
Student Management	Maintains student personal and academic records.	Add/edit student, program, year, status
Course Management	Creates and maintains the course catalogue.	Course code, credits, prerequisites, capacity
Enrollment Management	Core module for enroll, drop and approval.	Validation, waitlist, approval workflow
Faculty Management	Assigns faculty to courses and shows rosters.	Course allocation, class list export
Timetable Module	Prevents schedule conflicts.	Slot mapping, clash detection
Reports & Analytics	Provides insights for management.	Seat utilisation, demand, enrollment trend
Notification Module	Keeps users informed.	Email/SMS alerts, in-app messages

Business Rules

- A student cannot enroll in the same course twice in one semester.
- Enrollment is allowed only between the opening and closing dates set by the admin.
- Total credits per semester must stay within the allowed minimum and maximum.
- If a course is full, the student is added to the waitlist in first-come order.
- Dropping a course automatically promotes the first waitlisted student.

6. Database Design

A normalised relational schema (up to 3NF) is used to avoid redundancy. The main tables and their important attributes are listed below.

Table	Important Columns	Keys
users	user_id, email, password_hash, role, created_at	PK: user_id
students	student_id, user_id, name, dob, program, year, status	PK: student_id; FK: user_id
faculty	faculty_id, user_id, name, department	PK: faculty_id; FK: user_id
courses	course_id, code, title, credits, capacity, semester, faculty_id	PK: course_id; FK: faculty_id
prerequisites	course_id, required_course_id	Composite PK
enrollments	enrollment_id, student_id, course_id, status, enrolled_on	PK: enrollment_id; FK: student_id, course_id
timetable	slot_id, course_id, day, start_time, end_time, room	PK: slot_id; FK: course_id

Relationships

- One student can have many enrollments; one course can have many enrollments (many-to-many resolved by the **enrollments** table).
- One faculty member can teach many courses; each course has one primary faculty.
- A course can have many prerequisites, stored in a self-referencing table.
- Enrollment status values: PENDING, CONFIRMED, WAITLISTED, DROPPED, REJECTED.

Sample SQL

```
SELECT c.code, c.title, c.capacity - COUNT(e.enrollment_id) AS seats_left
FROM courses c LEFT JOIN enrollments e
  ON e.course_id = c.course_id AND e.status = 'CONFIRMED'
GROUP BY c.course_id;
```

7. Enrollment Workflow

The enrollment process is fully automated. The steps below show what happens from the moment a student clicks *Enroll*.

Step	Action	System Check / Outcome
1	Student logs in and searches for a course	Session and role verified
2	Student selects course and clicks Enroll	Enrollment window must be open
3	Prerequisite check	Reject with message if not completed
4	Timetable clash check	Reject if slot overlaps an enrolled course
5	Credit limit check	Reject if semester limit exceeded
6	Seat availability check	Seat free: CONFIRMED / Full: WAITLISTED
7	Optional admin approval	Status changes to CONFIRMED or REJECTED
8	Notification sent	Student receives email/SMS confirmation

Drop Process

When a student drops a course before the deadline, the record status becomes DROPPED and the seat count is updated immediately. The system then promotes the first waitlisted student, re-runs the validation checks for that student and sends a notification. All actions are stored in an audit log with timestamp and user ID for accountability.

Exception Handling

- Concurrent requests for the last seat are handled using database transactions and row locking.
- Network failures during enrollment roll back the transaction so no partial record is stored.
- Admins can override rules with a mandatory reason, which is written to the audit log.

8. User Interface & Testing

8.1 Main Screens

Screen	Users	Description
Login / Register	All	Secure access with role detection
Student Dashboard	Student	Enrolled courses, credits, notifications
Course Catalogue	Student	Search, filter, view seats and Enroll button
My Timetable	Student, Faculty	Weekly schedule view
Course Manager	Admin	Add / edit courses, set capacity and prerequisites
Enrollment Approvals	Admin	Approve, reject or override requests
Reports	Admin	Charts and export to PDF / Excel

8.2 Test Cases

ID	Test Scenario	Expected Result	Status
TC1	Login with valid credentials	Dashboard opens	Pass
TC2	Login with wrong password	Error message shown	Pass
TC3	Enroll in an open course	Status CONFIRMED	Pass
TC4	Enroll in a full course	Status WAITLISTED	Pass
TC5	Enroll without prerequisite	Enrollment rejected	Pass
TC6	Enroll in clashing timetable slot	Clash warning shown	Pass
TC7	Drop a course with waitlist	Next student promoted	Pass
TC8	Student opens admin URL	Access denied (403)	Pass

Unit testing covered individual validation rules, integration testing verified module interaction, and system testing confirmed the complete workflow under load.

9. Conclusion & Future Scope

9.1 Conclusion

The Student Enrollment & Course Management System replaces manual, error-prone procedures with an automated, transparent and secure platform. It reduces administrative workload, prevents seat and timetable conflicts, and gives students a convenient way to manage their academic path. The modular design and normalised database make the system reliable and easy to extend.

9.2 Advantages

- Saves time for students, faculty and administrative staff.
- Real-time seat tracking and automatic waitlist management.
- Improved data accuracy and complete audit trail.
- Data-driven reports for planning courses and capacity.

9.3 Future Enhancements

- Integration with online fee payment and examination modules.
- Mobile application for Android and iOS.
- AI-based course recommendation using a student's academic history.
- Single sign-on and integration with the institution's LMS.
- Advanced analytics dashboards with predictive demand forecasting.

9.4 References

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